Is corporatism green or dirty? Examining the effects of corporatism on climate policy

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Outline

Motivation

Theory

Empirics

Conclusion and outlook

Empirical motivation: correlational, cross-sectional evidence

Empirical literature

- Corporatism positively correlated with either emissions reductions or environmental (rather than climate) policy indicators (Scruggs, 1999, 2001, 2003), though not undisputed (Jahn, 2016b)
- Positive large-N evidence echoed by a fair amount of subsequent qualitative studies (Andersen, 2019; Gronow et al., 2019), though, again, not undisputed (Mildenberger, 2020)
  $\rightarrow$ same for studies on corporatism and long-term policymaking more generally (Jacobs, 2011; Lindvall, 2017; Seidl, 2023)
- Shortcomings: Samples end before "ratcheting up" of climate policy, emissions might not the theoretically most interesting DV, and no causal identification

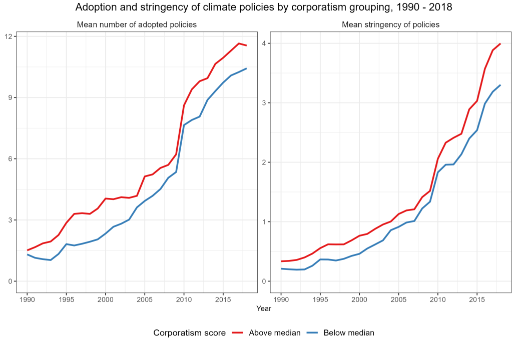

Figure: Evolution of climate policy adoption and stringency by corporatism score

▸ Descriptives I ▸ Descriptives II ▸ Descriptives III ▸ Descriptives IV

Empirical motivation: Cross-sectional and temporal variation

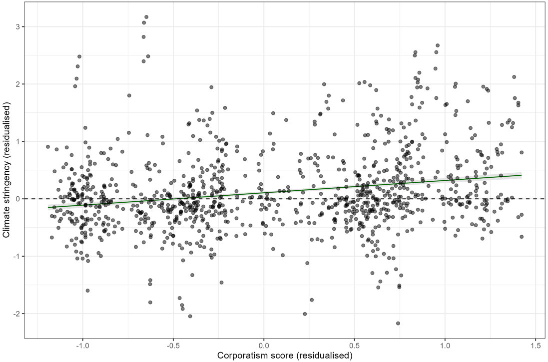

Figure: Cross-sectional variation

Notes: The residualised scatterplot visualises the variation that remains after controlling for year fixed effects, type of climate policy, and CO2 per capita. Standard errors are robust (Abadie et al., 2023).

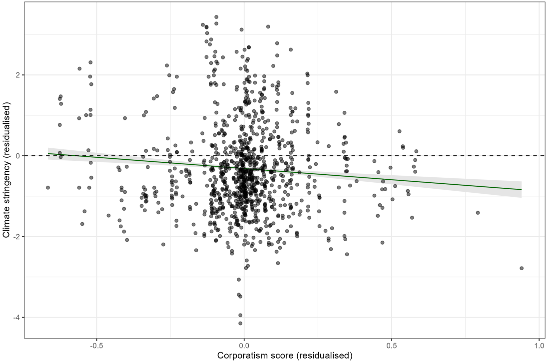

Figure: Temporal variation

Notes: The residualised scatterplot visualises the variation that remains after controlling for country fixed effects, type of climate policy, and CO2 per capita. Standard errors are robust (Abadie et al., 2023).

Theoretical motivation: Two contrasting perspectives

Corporatism facilitates climate policy

- Finnegan, 2022: Institutionalised tripartite concertation implies repeated interactions, which allow for credible promises of commitment (folk-theorem logic).
- While not quite the same, this view is similar to the Olsonian conception of corporatism (Boix, 1998; Garrett, 1998; Olson, 1982).
- Unclear under what conditions this effect materialises, especially when declines of union power might reduce the degree to which they are *encompassing* institutions.

Corporatism undermines climate policy

- Mildenberger, 2020: Concertation allows for 'double representation' of carbon-intensive producers' interests ('dirty' labour and capital), giving them veto power.
- They use that veto power in ways that are detrimental to climate policy.
- Unclear why no side payments exist that allow pro-climate interests to compensate carbon-intensive capital and labour (does the effect actually exist?), and under what conditions the effect is plausible.

▸ Literature, Part I ▸ Literature, Part II

Outline

Motivation

Theory

Empirics

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Towards reconciling these two perspectives

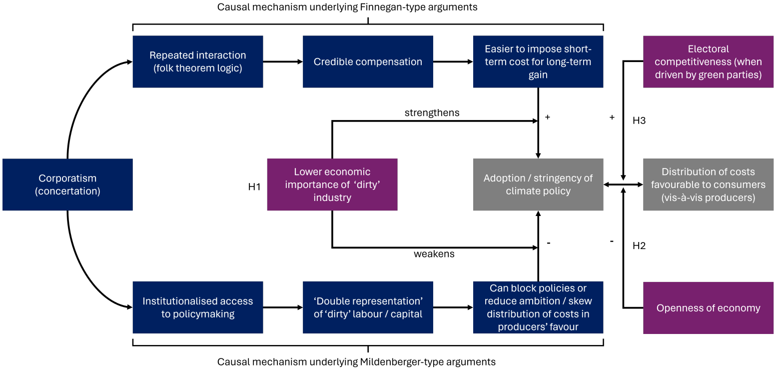

Figure: Theoretical framework

Notes: Purple boxes represent moderating variables, whilst grey boxes refer to my dependent variables of interest. The other boxes are merely coloured for emphasis. Finally, note that indirect effects are not visualised here for simplicity's sake.

Outline

Motivation

Theory

Empirics

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Inferential challenges and estimation

Two central challenges:

1. Corporatism is not randomly assigned
2. Corporatism exhibits relatively little variation over time

Cannot claim causal identification, but temporal granularity of data allows for more demanding FE specifications than the ones usually estimated

Other limitations: relatively little variation limits statistical power and data do not allow for inferences about mechanisms

$$Y_{it} = \beta_1 C_{it-1} + \beta_2 M_{it-1} + \beta_3 C_{it-1} \times M_{it-1} + \zeta \mathbf{X}^T_{it-1} + \eta_i + \gamma_{t(k)} + \epsilon_{it}$$

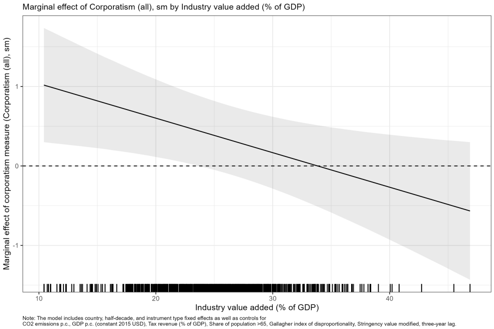
Hypothesis	**Dependent variable**	**Moderating variable**	**Expected sign for β_3**
H1	Costs for consumers	Economic importance of carbon-intensive industry	$\beta_3 > 0$
	Policy stringency		$\beta_3 < 0$
H2	Costs for consumers	Openness of economy	$\beta_3 > 0$
	Policy stringency		ambiguous
H3	Costs for consumers	Electoral competitiveness	$\beta_3 < 0$
	Policy stringency	Competition by green parties	$\beta_3 > 0$

Table: Translating the theoretical hypotheses into expected parameter signs

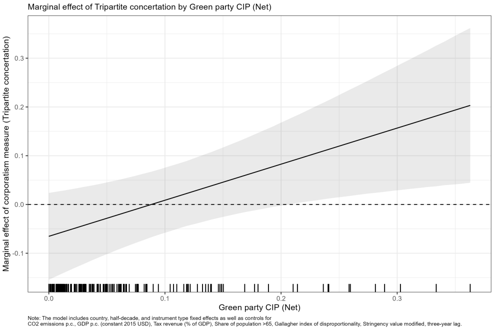
Variables and their operationalisation

Variable	Operationalisation	Data source(s)
Dependent variables		
Climate policy stringency	Ambition level, relative to all other countries in a given year	Stringency variable, OECD CAPMF database (Nachtigall et al., 2024)
	Overall costs of climate policy	Finnegan, 2022
Distribution of costs between consumers and producers	Shadow carbon prices for consumers and producers	Althammer and Hille, 2016; Finnegan, 2022
Independent variable		
Corporatism	(Smoothed) corporatism index	Jahn's time-varying index (Jahn, 2016a,b)
	Tripartite concertation dummy	ICTWSS database
	Concertation index	Finnegan, 2022
Moderating variables		
Economic importance of carbon-intensive industry	% of GDP/value added	CPDS (Armingeon et al., 2023), WDI
Electoral competitiveness	Probability of losing/winning office	Kayser and Lindstädt, 2015
	Coalition inclusion probability by green party/parties	Kayser, Orlowski, and Rehmert, 2023; Kayser and Rehmert, 2021
Openness of the economy	Total trade as % of GDP	CPDS (Armingeon et al., 2023), WDI
	Trade CO_2 share	OWID

Stringency-related results

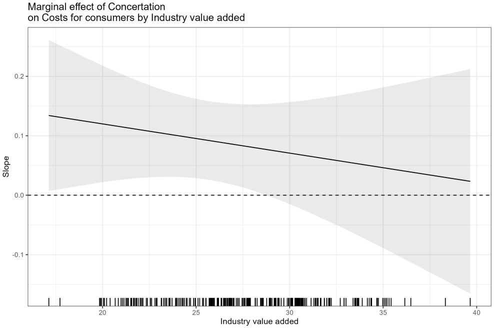


((a)) Marginal 'effect' of corporatism by industry value added (H1)

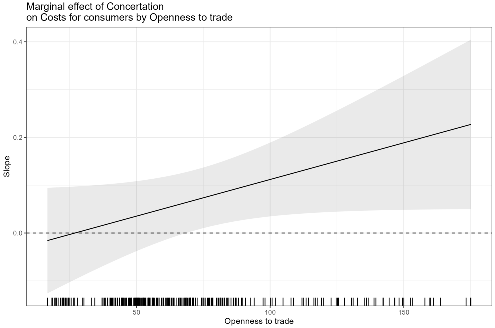


((b)) Marginal 'effect' of tripartite by net green party CIP (H3)

Distribution of the Costs of Climate Policy (I)

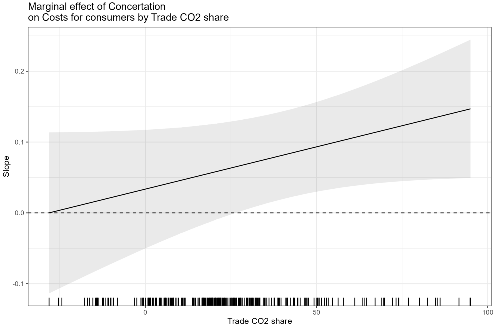


((a)) Marginal ’effect’ of concertation by industry value added (H1)

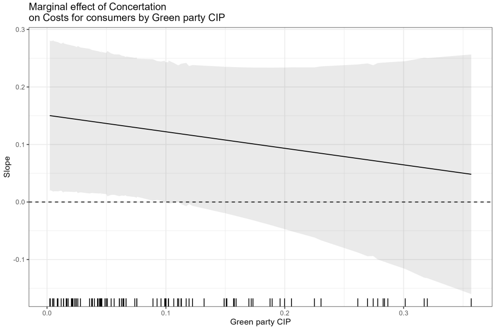


((b)) Marginal ’effect’ of concertation by openness of economy (H2)

Distribution of the Costs of Climate Policy (II)



((a)) Marginal 'effect' of concertation by CO_2 emissions embedded in trade (H2)



((b)) Marginal 'effect' of concertation by green party CIP (H3)

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Conclusion, roads not yet taken, and open questions

Theoretical argument: Both Finnegan and Mildenberger are two special cases (if at all). The former neglects the importance of electoral competition; the latter the ability of corporatist actors to internalise negative externalities.

Empirical results: Support for the stringency components of H1 and H3 (no clear sign predicted for H2); for the distribution of costs, the signs are as expected, but only clear support for H2.

Plan: Currently working on improving the theoretical framework and will then turn to improving the empirics

Open questions: What should be the main focus for empirical improvements? Reduced-form? Mechanisms?

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Outline

Descriptives on climate policy and corporatism

Climate policy

Corporatism

Institutional turn in the climate politics literature

Adoption and stringency over time

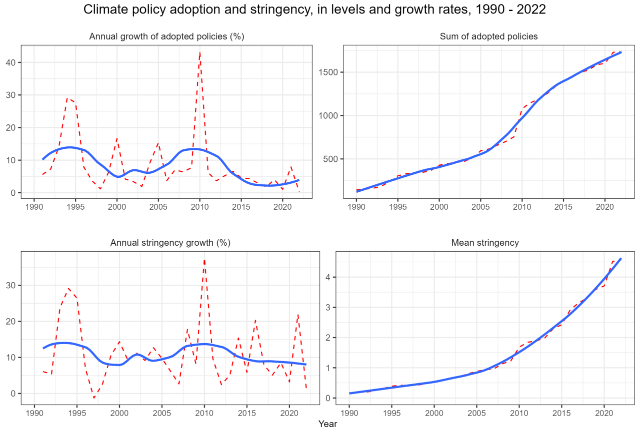

Figure: Evolution of level and growth rate in climate policies adopted and their stringency

Back

Adoption and stringency over time and across sectors

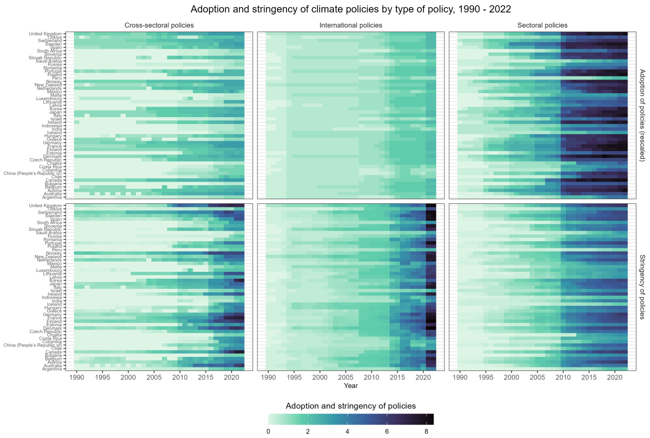

Figure: Heatmap of climate policy adoption and stringency over time, by level-one policy type

Adoption and stringency over time, across sectors, and by instrument type

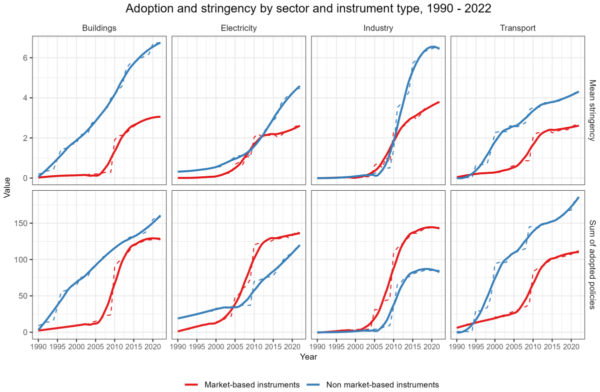

Figure: Evolution of adoption and stringency over time, by instrument type and sector

Back

Variation in corporatism over time

Figure: Changes in corporatism scores by country and half-decade

Back

Outline

Descriptives on climate policy and corporatism

- Climate policy
- Corporatism

Institutional turn in the climate politics literature

Macro-political institutions in the climate politics literature (Part 1)

Macro-political institution	Explanation (taken from Zwar, Edenhofer, and Flachsland, 2024)
Electoral rules (Bormann and Golder, 2022)	Several studies (Finnegan, 2022; K. Harrison, 2010; K. Harrison and Sundstrom, 2010; Lockwood et al., 2017) identify electoral rules—the rules for translating votes into seats—as an important determinant of climate policy. Most studies argue that proportional representation (PR), compared to majoritarian, tends to result in more stringent climate policies (Meckling and Karplus, 2023).
Interest group mediation (Lijphart, 2012; Siaroff, 1999)	Patterns of interest group mediation appear to affect climate policy (Finnegan, 2022; Jahn, 2016b; Mildenberger, 2020; Scruggs, 1999, 2003). Corporatist arrangements, some argue, facilitate long-term climate policymaking by enabling governments to commit credibly to compensation (Finnegan, 2022). Others, however, believe corporatism to be detrimental due to the 'double representation' of carbon-intensive workers and businesses (Mildenberger, 2020).
Type of welfare state (Esping-Andersen, 1990)	Extensive welfare states, compared to minimal ones, allow governments to compensate for the losers of climate policy more effectively and are considered crucial for the transition to a low-carbon economy (Gough, 2016; Meckling, Lipscy, et al., 2022). Other studies focus on the relationship between welfare states and climate policy features, such as citizens' preferences for climate instruments (Sivonen and Kukkonen, 2021).

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Macro-political institutions in the climate politics literature (Part 2)

Macro-political institution	Explanation (taken from Zwar, Edenhofer, and Flachsland, 2024)
Type of economy (Hall and Soskice, 2001)	Hall and Soskice (Hall and Soskice, 2001) argue that modern capitalist arrangements are split into liberal market and coordinated market economies (LMEs and CMEs). Studies examine how these influence distinct decarbonization methods (Ćetković et al., 2016), including phasing out coal (Rentier, Lelieveldt, and Kramer, 2019), green innovation (Mikler and N. E. Harrison, 2012), and the limits of LMEs in environmental taxes (MacNeil, 2016).
Structure of state	The state structure is hypothesized to affect climate policy through efficient vertical coordination in unitary states and policy experimentation in federal systems (Callander and Harstad, 2015; Reich, 2021). Research highlights coordination as vital, especially in polarised contexts like the US, with sub-national experimentation aiding stringent climate policies (Christoff and Eckersley, 2021; Karapin, 2020; Mildenberger, 2021; Müller and Slominski, 2017; Trachtman, 2019).
EU membership	EU membership has influenced climate policy outcomes (Bayer and Aklin, 2020; Calel and Dechezleprêtre, 2016; Dechezleprêtre, Nachtigall, and Venmans, 2023; Eskander and Fankhauser, 2020), particularly through policy diffusion and convergence (Gawel and Strunz, 2019; Grafström et al., 2023; Strunz et al., 2018).

Back